Type MS · Turn a flowchart or pseudocode into Python

Answer 1 of 8 · Question 3 part c · 10 marks

2023 JUN · Paper 02 · extract 1 of 6 · find part c in this mark scheme extract

3(a)	A1 A2	Award **one** mark for any of the following up to a maximum of **two** marks: • Each column represents a variable/output (1) • Rows shows how the values of variables change (as the code is executed) (1) • Values are filled in manually/by hand (1)	Line numbers are for guidance – mark answer as a whole	**2**

Question	mp	Answer	Additional Guidance	Mark
3(b)	B1 B2 B3	Award **one** mark for each correct cell: **num1** | **num2** | **Output** 88 | 18 | State 2 17 | 18 | State 4 12 | 19 | State 1	ignore quotes e.g. "State 2"	**3**

Type MS · Turn a flowchart or pseudocode into Python

Answer 1 of 8 · Question 3 part c · 10 marks

2023 JUN · Paper 02 · extract 2 of 6 · find part c in this mark scheme extract

Question	mp	Answer	Additional guidance	Mark
3(c)	Award **one** mark for each of:		C4 should exit loop Allow C5 if a loop is used to keep asking for a non-negative exponent C9 – ignore error message in C6; must have base, exponent, and answer in output for this mark	**10**
	C1	At least one variable with a suitable variable name		
	C2	At least one variable initialised to an appropriate value		
	C3	While loop used to continue running code		
	C4	Relational test to identify terminating condition for loop		
	C5	Selection used to check for negative exponent		
	C6	Appropriate error message if exponent is negative		
	C7	Calculation of answer using any method		
	C8	A loop used to calculate answer as indicated in flowchart		
	C9	Prompts and output messages are fit for purpose		
	C10	Executing and producing the correct output for anticipated inputs		

Suggested test data and output		
Base	**Exponent**	**Output**
0		ends program
2	0	2 to power of 0 is 1
3	4	3 to power of 4 is 81

Type MS · Turn a flowchart or pseudocode into Python

Answer 1 of 8 · Question 3 part c · 10 marks

2023 JUN · Paper 02 · extract 3 of 6 · find part c in this mark scheme extract

C#

```
// Initialise variables
int baseNum = 0;
int exponent = 0;
int answer = 1;
String outString = "";

// Display the first prompt and get the base number
Console.Write("Enter a base number (0 to exit)): ");
baseNum = Convert.ToInt32(Console.ReadLine());

// Check if user wants to exit
while (baseNum != 0.0)
{
    // Display a prompt and get the exponent number
    Console.Write("Enter an exponent number: ");
    exponent = Convert.ToInt32(Console.ReadLine());

    // Check if the exponent is negative
    if (exponent < 0)
    {
        Console.Write("Invalid exponent entered");
    }
    else
    {
        answer = 1;

        // Calculate the answer using exponentiation
        for (int i = 0; i < exponent; i++)
        {
            answer = answer * baseNum;
        }

        // Create a string for the output and show the user
        outString = baseNum.ToString() + " to the power of " +
                    exponent.ToString() + " is " +
                    answer.ToString();
        Console.Write(outString);
    }
    // Display the first prompt again and get the base number
    Console.Write("\nEnter a base number (0 to exit)): ");
    baseNum = Convert.ToInt32(Console.ReadLine());
}
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 1 of 8 · Question 3 part c · 10 marks

2023 JUN · Paper 02 · extract 4 of 6 · find part c in this mark scheme extract

Java

```
// --------------------------------------------------
// Write your code below this line

// Initialise variables
int baseNum = 0;
int exponent = 0;
int answer = 1;
String outString = "";

Scanner myScanner = new Scanner (System.in);

// Display the first prompt and get the base number
System.out.print("Enter a base number (0 to exit)): ");
baseNum = Integer.parseInt (myScanner.nextLine());

// Check if user wants to exit
while (baseNum != 0.0)
{
    // Display a prompt and get the exponent number
    System.out.print("Enter an exponent number: ");
    exponent = Integer.parseInt (myScanner.nextLine());

    // Check if the exponent is negative
    if (exponent < 0)
    {
        System.out.println ("Invalid exponent entered");
    }
    else
    {
        answer = 1;

        // Calculate the answer using exponentiation
        for (int i=0; i<exponent; i++)
        {
            answer = answer * baseNum;
        }

        // Create a string for the output and show the user
        outString = Integer.toString(baseNum) + " to the power of " +
                    Integer.toString(exponent) + " is " +
                    Integer.toString(answer);
        System.out.println(outString);
    }

    // Display the first prompt again and get the base number
    System.out.print("Enter a base number (0 to exit)): ");
    baseNum = Integer.parseInt(myScanner.nextLine());
}
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 1 of 8 · Question 3 part c · 10 marks

2023 JUN · Paper 02 · extract 5 of 6 · find part c in this mark scheme extract

Python

```
# Initialise variables
base = 0
exponent = 0
answer = 1

# Get the first base number
base = int (input ("Enter a base number (0 to exit): "))

# Check if the user wants to exit
while (base != 0):

    # Get the exponent number
    exponent = int (input ("Enter an exponent number: "))

    # Check if the exponent is negative
    if (exponent < 0):
        print ("Invalid exponent entered")
    else:
        # Reset the answer
        answer = 1

        # Calculate the answer using exponentiation
        for count in range (0, exponent):
            answer = answer * base

        # Create a string for the output sentence
        print (str (base) + " to the power of " +
               str (exponent) + " is " +
               str (answer))

    # Get another base number and lp
    base = int (input ("Enter a base number (0 to exit): "))
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 1 of 8 · Question 3 part c · 10 marks

2023 JUN · Paper 02 · extract 6 of 6 · find part c in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Turn a flowchart or pseudocode into Python

Answer 2 of 8 · Question 2 part a · 10 marks

2019 JUN · Paper 02 · extract 1 of 13 · find part a in this mark scheme extract

2(b)(i)	24.0 (1)	Accept 24	(1)

Type MS · Turn a flowchart or pseudocode into Python

Answer 2 of 8 · Question 2 part a · 10 marks

2019 JUN · Paper 02 · extract 2 of 13 · find part a in this mark scheme extract

2(b)(ii)	One of: By the rules of precedence multiplication is done before addition. (1) Without the brackets the addition would be done to the total of the multiplication. (1) By the rules of the BODMAS/BIDMAS additions happen after multiplication. (1) newSpeed and intialSpeed have to be added together before multiplying (by time). (1)		(1)

Type MS · Turn a flowchart or pseudocode into Python

Answer 2 of 8 · Question 2 part a · 10 marks

2019 JUN · Paper 02 · extract 3 of 13 · find part a in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Turn a flowchart or pseudocode into Python

Answer 2 of 8 · Question 2 part a · 10 marks

2019 JUN · Paper 02 · extract 4 of 13 · find part a in this mark scheme extract

2(b)(iii)	Real (1) Because real x integer = real (1)	Accept double / float Accept 'because it could be a number with decimal places'	(2)

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Answer 2 of 8 · Question 2 part a · 10 marks

2019 JUN · Paper 02 · extract 5 of 13 · find part a in this mark scheme extract

2(b)(iv)	27.375 (1)		(1)

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Answer 2 of 8 · Question 2 part a · 10 marks

2019 JUN · Paper 02 · extract 6 of 13 · find part a in this mark scheme extract

Q2(c)(i)	C#: 16 or 20 (1) Java: 16 or 21 (1) Python: 8 or 10 (1)		(1)

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Answer 2 of 8 · Question 2 part a · 10 marks

2019 JUN · Paper 02 · extract 7 of 13 · find part a in this mark scheme extract

Q2(c)(ii)	C#: 29 or 31 (1) Java: 24 or 26 (1) Python: 14 or 15 (1)		(1)

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Answer 2 of 8 · Question 2 part a · 10 marks

2019 JUN · Paper 02 · extract 8 of 13 · find part a in this mark scheme extract

Q2(c)(iii)	Any of: 1. Message (1) 2. startPos (Python) / pos (Java or C#) (1) 3. characters (1)		(1)

Type MS · Turn a flowchart or pseudocode into Python

Answer 2 of 8 · Question 2 part a · 10 marks

2019 JUN · Paper 02 · extract 9 of 13 · find part a in this mark scheme extract

Q2(c)(iv)	newMessage (1)	Do not accept outMessage	(1)

Type MS · Turn a flowchart or pseudocode into Python

Answer 2 of 8 · Question 2 part a · 10 marks

2019 JUN · Paper 02 · extract 10 of 13 · find part a in this mark scheme extract

Q2(c)(v)	Response starts with 'c' (1) Response has six letters in sequence from 'educational' (1)	Marks can be awarded from Q2(c)(vi)	(2)

Type MS · Turn a flowchart or pseudocode into Python

Answer 2 of 8 · Question 2 part a · 10 marks

2019 JUN · Paper 02 · extract 11 of 13 · find part a in this mark scheme extract

Q2(c)(vi)	1. Returns substring / crops a string (1) 2. Located between the two input values / with number of characters identified in the last parameter (characters) and starting at the second parameter (startPos) (1)	Award two marks for: It produces a substring of the message starting at position startPos with length characters Award up to 2 marks for a correct interpretation of the code (e.g 'obtain a newMessage' – is worth 1) Accept a correct generic explanation of what a subprogram does for 1 mark	(2)

Type MS · Turn a flowchart or pseudocode into Python

Answer 2 of 8 · Question 2 part a · 10 marks

2019 JUN · Paper 02 · extract 12 of 13 · find part a in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · Turn a flowchart or pseudocode into Python

Answer 2 of 8 · Question 2 part a · 10 marks

2019 JUN · Paper 02 · extract 13 of 13 · find part a in this mark scheme extract

2(a)	Award 1 mark for each of: 1. set counter to 1 and guess to 0 (1) 2. set answer to random value between 1 and 10 (1) 3. request input of and accept an integer value for guess (1) 4. create a while loop with a correct condition (1) 5. increment the counter (1) 6. use if else selection to determine and display appropriate output message for incorrect guesses (1) 7. request input of and accept another integer value for a guess inside the loop (1) 8. display correct guess and count of guesses (1) 9. compiling without syntax errors (1) 10. coding meets all requirements of question (1)	Candidates are required to open the file Q02a in the code editor. Amended code should be saved as Q02aFINISHED. Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements.	(10)

Question	Answer	Additional Guidance	Mark

Type MS · Turn a flowchart or pseudocode into Python

Answer 3 of 8 · Question 2 part a · 10 marks

2020 NOV · Paper 02 · extract 1 of 7 · find part a in this mark scheme extract

2 (a)	A1	2.3.4	Set variable base to 50 or heightChk to true	Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements. Don't penalise spelling mistakes and alternative wording of the output.	(10)
	A2	2.2.2	Create WHILE DO loop		
	A3	2.4.1	Request input of height		
	A4	2.2.2	if statement checks if height >= 1 or <= 100		
	A5	2.2.2	if statement checks if height is >= 1 and <=100		
	A6	2.2.2	heightCheck set to false if condition is met		
	A7	2.5.1	Set area using given formula (0.5 x base x height)		
	A8	2.4.1	Display one of base, height and area with appropriate label or displays two of base, height and area without labels		
	A9	2.4.1	Display all of base, height and area with appropriate labels		
	A10	1.1.6	Executing and producing correct output		
Code example					
Python					

```python
#   Q02aFINISHED

#   Initialise variables

base = 50

#   Print prompt and take input from user

    heightChk = TRUE
    while heightChk:
        height = int(input("Enter the height (between 1 and 100): "))
        if (height >= 1 and height <= 100):
            heightChk = FALSE

#   Calculate area and print out values

    area = 0.5 * base * height

    print ("Base length of triangle is : ", base)
    print ("Height of triangle is : ", height)
    print ("Area of triangle is : ", area)
```

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Type MS · Turn a flowchart or pseudocode into Python

Answer 3 of 8 · Question 2 part a · 10 marks

2020 NOV · Paper 02 · extract 2 of 7 · find part a in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Turn a flowchart or pseudocode into Python

Answer 3 of 8 · Question 2 part a · 10 marks

2020 NOV · Paper 02 · extract 3 of 7 · find part a in this mark scheme extract

2 (b)	B1	2.2.1	Meaningful variable names used (1)	The code example shown is one way of responding to the task. Other methods should be credited accordingly.	(7)
	B2	2.4.1	Meaningful prompts for input (1)		
	B3	2.4.1	Input of length and width as whole numbers (1)		
	B4	2.5.1	Perimeter calculated (= 2 x length + 2 x width) (1)		
	B5	2.5.1	Gap deducted from perimeter (1)		
	B6	2.4.1	Meaningful output (1)		
	B7	2.1.1	Code is fit for purpose. Must have mark points 1-6 and be easy to read (1)		

Code example

Python

```
1   #   Q02bFINISHED
2
3   #   Request input
4
5   length = int(input("Enter the length: "))
6   width = int(input("Enter the width: "))
7
8   #   Calculate number of panels needed
9
10  panels = 2 * length + 2 * width - 4
11
12  #   Print out number of panels needed
13
14  print ("Number of panels needed: " , panels)
```

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Type MS · Turn a flowchart or pseudocode into Python

Answer 3 of 8 · Question 2 part a · 10 marks

2020 NOV · Paper 02 · extract 4 of 7 · find part a in this mark scheme extract

Question	mp	ref	Answer	Additional Guidance	Mark

Type MS · Turn a flowchart or pseudocode into Python

Answer 3 of 8 · Question 2 part a · 10 marks

2020 NOV · Paper 02 · extract 5 of 7 · find part a in this mark scheme extract

For Q6, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit.

Question	mp	ref	Answer	Additional Guidance	Mark
6	A1	2.2.1	Initialise variable for total sales by member of staff (1)	Sales for each member of staff George Taylor 28466 Fehlix Chayne 35250 Sumren Bergen 30186 Samira Beckle 37430 Nellie Bogart 20448 Cheryl Grouth 34383 Danuta Graunt 26421 Jaiden Deckle 21507 Jimran Caliks 32415 Deynar Derran 28590 ========== Total staff sales: 295096 Highest sales by: Samira Beckle with 37430 Second highest sales by: Fehlix Chayne with 35250 >>>	(11)
	A2	2.5.1	Add succeeding months' sales to total sales (1)		
	A3	2.1.6	Display total sales by member of staff (1)		
	A4	2.2.2	Loop repeats for each member of staff (1)		
	A5	2.2.1	Running team total sales initialised (1)		
	A6	2.2.2	Add succeeding individual staff sales to team total (1)		
	A7	2.1.6	Display final running total (1)		
	A8	2.1.6	Staff member with highest individual sales identified (1)		
	A9	2.1.6	Staff member with second highest individual sales identified (1)		
	A10	2.1.6	Information about staff with highest sales displayed (1)		
	A11	2.1.6	Information about staff with second highest sales displayed (1)		

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Type MS · Turn a flowchart or pseudocode into Python

Answer 3 of 8 · Question 2 part a · 10 marks

2020 NOV · Paper 02 · extract 6 of 7 · find part a in this mark scheme extract

Band 1 (1-3 marks)	Band 2 (4-6 marks)	Band 3 (7-9 marks)	Mark
Little attempt to decompose the problem into component parts	Some attempt to decompose the problem into component parts	The problem has been decomposed into component parts	
Some parts of the logic are clear and appropriate to the problem	Most parts of the logic are clear and mostly appropriate to the problem	The logic is clear and appropriate to the problem	
Some appropriate use and manipulation of data types, variables, data structures and program constructs	The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate	The use and manipulation of data types, variables and data structures and program constructs is appropriate	
Parts of the code are clear and readable	Code is mostly clear and readable	Code is clear and readable	
Finished program will not be flexible enough with other data sets or input	Finished program will function with some but not all other data sets or input	Finished program could be used with other data sets or input	
The program meets some of the given requirements	The program meets most of the given requirements	The program fully meets the given requirements	(9)

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Type MS · Turn a flowchart or pseudocode into Python

Answer 3 of 8 · Question 2 part a · 10 marks

2020 NOV · Paper 02 · extract 7 of 7 · find part a in this mark scheme extract

Code example

Python

```
allStaffTotal = 0
highestSales = 0
secondSales = 0
high = 0
second = 0

print ("Sales for each member of staff","\n\n")
for staff in staffSales:
    staffTotal = 0
    sales = 3
    while (sales < 9):
        staffTotal = staffTotal + staff[sales]
        sales = sales + 1
    print (staff[1],staff[2],staffTotal)
    if (staffTotal > secondSales):
        if (staffTotal > highestSales):
            highestSales = staffTotal
            high = staff
        else:
            secondSales = staffTotal
            second = staff
    allStaffTotal = allStaffTotal + staffTotal
print ("==========\n\nTotal sales for all staff: ", allStaffTotal)

print ("\n\nHighest sales by: ",high[1],high[2], " with ", highestSales)
print ("\n\nSecond highest sales by: ",second[1],second[2], " with ", secondSales)
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 4 of 8 · Question 2 part a · 11 marks

2021 JUN · Paper 02 · extract 1 of 6 · find part a in this mark scheme extract

<table>
<tr><th>Question</th><th>Mp</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>2 (c)</td><td colspan="2">Award 1 mark for each set of test data.</td><td></td><td></td></tr>
<tr><td></td><td>C1
C2
C3</td><td>
<table>
<tr><th></th><th>Test data</th><th>Expected results</th></tr>
<tr><td>booksSold
profit</td><td>Either of
• booksSold = 4
• profit = 4</td><td>Poor performances this week</td></tr>
<tr><td>booksSold</td><td>5</td><td rowspan="2">Sales and profit are good this week</td></tr>
<tr><td>profit</td><td>10</td></tr>
<tr><td>booksSold</td><td>21</td><td rowspan="2">Sales and profit are excellent this week</td></tr>
<tr><td>profit</td><td>20</td></tr>
</table>
</td><td></td><td>(3)</td></tr>
</table>

Type MS · Turn a flowchart or pseudocode into Python

Answer 4 of 8 · Question 2 part a · 11 marks

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Question	mp	Answer	Additional Guidance	Mark

Type MS · Turn a flowchart or pseudocode into Python

Answer 4 of 8 · Question 2 part a · 11 marks

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Question		Answer	Additional Guidance	Mark
2 (a)	Award one mark for each of:		Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements. Do not penalise spelling mistakes in the assignment of username and/or password Do not penalise spelling mistakes and alternative wording of the output.	(11)
	A1	At least one variable with a suitable variable name		
	A2	username = bard423		
	A3	password = nX2934?		
	A4	Loop used		
	A5	Username or password entered		
	A6	Username or password stored in variable(s)		
	A7	At least one suitable input message		
	A8	Checks username and password		
	A9	Appropriate error message(s) displays		
	A10	Welcome message displayed		
	A11	Executing and producing correct output (must have A4)		
Code example				
Python				

```
# Initialise variables
username = "bard423"
password = "nX2934?"
count = 0
inputUsername = ""
inputPassword = ""

# Print promts, take and check input from user
while inputUsername != username or inputPassword != password:
    if count > 0:
        print("There is a problem with the login details. Try again")
    count = count + 1
    inputUsername = input("Enter your username ")
    inputPassword = input("Enter your password ")
print("Welcome")
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 4 of 8 · Question 2 part a · 11 marks

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Type MS · Turn a flowchart or pseudocode into Python

Answer 4 of 8 · Question 2 part a · 11 marks

2021 JUN · Paper 02 · extract 5 of 6 · find part a in this mark scheme extract

Question	Mp	Answer		Additional Guidance	Mark
2 (b)	Award 1 mark for each correct condition.			Accept alternatives e.g. Line 11 booksSold <=4 etc.	(4)
		Condition	**Output message**		
	B1	Number of books sold is under 5 or profit made is under 5	Poor performance this week		
	B2	Number of books sold is over 20; profit made is at least 20	Sales and profit are excellent this week		
	B3	Number of books sold is at least 5; profit made is at least 10	Sales and profit are good this week		
	B4	All other inputs	Alert manager		
Code example					
Python					

```
11 if booksSold < 5 or profit < 5:
12     print("Poor performance this week")
13 elif booksSold > 20 and profit >= 20:
14     print("Sales and profit are excellent this week")
15 elif booksSold >=5 and profit >= 10:
16     print("Sales and profit are good this week")
17 else:
18     print("Alert manager")
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 4 of 8 · Question 2 part a · 11 marks

2021 JUN · Paper 02 · extract 6 of 6 · find part a in this mark scheme extract

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

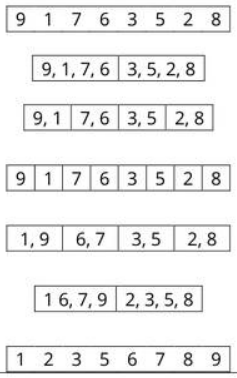
2021 NOV · Paper 02 · extract 1 of 18 · find part a in this mark scheme extract

2 (b)	B1 B2 B3 B4	**Award up to 4 marks for a linked explanation such as:** • The list is not sorted (1) Juan should come after Elija if the list was sorted / Elija is compared with Juan (1), so the bottom half of the list would be discarded (1) after the first pass through the loop (1) • Binary search works on a sorted list (1) because it uses divide and conquer (1) half the list discarded each pass through (1) based on comparison of search item with middle item (1)		(4)
Question	**mp**	**Answer**	**Additional Guidance**	**Mark**

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 2 of 18 · find part a in this mark scheme extract

2 (c)	C1 C2 C3 C4 C5	**Award 1 mark for each of:** Split into two sets of four (1) Split each set into two pairs (1) Split each pair into single elements (1) Merge elements into sorted pairs (1) Merge pairs into sorted sets (1) 	Allow step at C3 to be implied	(5)

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

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Question	mp	Answer	Additional Guidance	Mark

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 4 of 18 · find part a in this mark scheme extract

2 (a)	**Award one mark for each of:**		Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements. Do not penalise spelling mistakes in the input message.	
	A1	Input as an integer (1)		
	A2	Loop with correct conditions start <= end and not found (1)		
	A3	Use of correct symbol for integer division (in conversion of DIV) or type coercion (1)		
	A4	Check to see if number in middle matches item (1)		
	A5	Set found to true (1)		
	A6	Check to see if item is less than number in middle (1)		
	A7	Set end to middle – 1 (1)		
	A8	else – set start to middle + 1 (1)		
	A9	count incremented by 1 (1)		
	A10	Program execution is fully correct (1)		(10)
Code examples				

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 5 of 18 · find part a in this mark scheme extract

C#

```
Console.WriteLine("What is the number to find? ");
int item = Convert.ToInt32(Console.ReadLine());

while (start <= end && !found)
{
    middle = (start + end) / 2;
    if (numberList[middle] == item)
    {
        found = true;
    }
    else
    {
        if (item < numberList[middle])
        {
            end = middle - 1;
        }
        else
        {
            start = middle + 1;
        }
    }
    count++;
}
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 6 of 18 · find part a in this mark scheme extract

Java

```
Scanner input = new Scanner(System.in);
System.out.print("What is the number to find? ");
int item = Integer.parseInt(input.nextLine());

while (start <= end && !found)
{
    middle = (start + end) / 2;
    if (numberList[middle] == item)
    {
        found = true;
    }
    else
    {
      if (item < numberList[middle])
      {
          end = middle - 1;
      }
      else
      {
          start = middle + 1;
      }
    }
    count ++;
}
```

Python

```
item = int(input("What is the number to find? "))
while start <= end and not found:
    middle = (start + end) // 2
    if numberList[middle] == item:
        found = True
    else:
        if item < numberList[middle]:
            end = middle - 1
        else:
            start = middle + 1
    count = count + 1
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 7 of 18 · find part a in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 8 of 18 · find part a in this mark scheme extract

Question	mp	Answer	Additional Guidance	Mark
4 (a)	**Award one mark for each of:**			
		Validation		
	A1	Suitable input prompt and storing it in binaryPattern (1)		
	A2	Length of binary pattern validated as 8 (1)		
	A3	Loop used to check the length of binary pattern (1)		
		Binary to denary		
	A4	Loop through each digit in the binary pattern (1)		
	A5	Correct conversion of at least one binary digit to relevant placeholder (1)		
	A6	Attempt at running total using denaryNumber (1)		
	A7	Correct running total using denaryNumber (1)		
	A8	Print statement includes the binary pattern and the denary number (1)		
	A9	Print statement outside of conversion loop (1)		
	A10	Comment explains how conversion works (1)		
	A11	Program executes correctly for any binary pattern that is exactly 8 characters long only (1)		(11)
Code examples				

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 9 of 18 · find part a in this mark scheme extract

C#

```
while (binaryPattern.Length != 8)
{
    Console.WriteLine("Enter an 8 digit binary number to convert to denary ");
    binaryPattern = Console.ReadLine();
}

foreach (char digit in binaryPattern)
{
    if (digit == '1')
    {
        denaryNumber += denaryPlaceholders[count] * 1;
    }
    count++;
}
Console.WriteLine(binaryPattern + " converted to denary is " + denaryNumber);
Console.ReadLine();
```

Java

```
while (binaryPattern.length() != 8)
{
    Scanner input = new Scanner(System.in);
    System.out.print("Enter an 8 digit binary number to convert to denary ");
    binaryPattern = input.next();
}

for (char digit : binaryPattern.toCharArray())
{
    if (digit == '1')
    {
       denaryNumber += denaryPlaceholders[count] * 1;
    }
    count ++;
}
System.out.print(binaryPattern + " converted to denary is " + denaryNumber);
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 10 of 18 · find part a in this mark scheme extract

Python

```
while len(binaryPattern) != 8:
    binaryPattern = input("Enter an 8 digit binary number to convert to denary: ")

for digit in binaryPattern:
    if digit == "1":
        denaryNumber = denaryNumber + denaryPlaceholders[count] * 1
    count = count + 1

print(binaryPattern,"converted to denary is", denaryNumber)
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

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For Q5, the first 11 marks are for coding that matches requirements of task. The remaining 9 marks should be allocated on a best fit

Question	mp	Answer	Additional Guidance	Mark
5	A1	At least three variables initialised appropriately		(1)
	A2	Word input		(1)
	A3	Program repeats until 1 is input		(1)
	A4	Loop through each word in the array		(1)
		Words that begin with the same letter		
	A5	Check to see if the first letter of the input word matches the first letter of at least one word in the array		(1)
	A6	The number of words that begin with the same letter displayed		(1)
		Words that contain the input word		
	A7	Check to see if the word contains the input word		(1)
	A8	Number of letters in the word calculated		(1)
	A9	At least one word from the array that contains the input word displayed		(1)
	A10	The number of words that contain the input word displayed		(1)
	A11	Identification of the longest word or the shortest word		(1)

Band 1 (1-3 marks)	Band 2 (4-6 marks)	Band 3 (7-9 marks)	Mark
Little attempt to decompose into component parts	Some attempt to decompose into component parts	The problem has been decomposed into component parts	
Some parts of the logic are clear and appropriate to the problem	Most parts of the logic are clear and mostly appropriate to the problem	The logic is clear and appropriate to the problem	
Some appropriate use and manipulation of data types, variables, data structures and program constructs	The use and manipulation of data types, variables and data structures and program constructs is mostly appropriate	The use and manipulation of data types, variables and data structures and program constructs is appropriate	
Parts of the code are clear and readable	Code is mostly clear and readable	Code is clear and readable	
Finished program will not be flexible enough with other data sets or input	Finished program will function with some but not all other data sets or input	Finished program could be used with other data sets or input	
The program meets some of the given requirements	The program meets most of the given requirements	The program fully meets the given requirements	(9)

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 12 of 18 · find part a in this mark scheme extract

Code examples

C#

```
// Add your code here
string inputWord = "";

while (inputWord != "1")
{
    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    Console.WriteLine("Enter a word or 1 to exit ");
    inputWord = Console.ReadLine();

    if (inputWord != "1")
    {
        int count = 0;

        foreach (String word in wordArray)
        {
            if(word[0] == inputWord[0])
            {
                Console.WriteLine(word);
                count ++;
            }
        } // End of loop

        Console.WriteLine(count + " word(s) beginning with "+ inputWord[0]);
        Console.WriteLine("----------------------------------------------------------------------");

        count = 0;

        foreach (String word in wordArray)
        {
            if(word.Contains(inputWord))
            {
                count ++;
                Console.WriteLine(word);

                int length = word.Length;

                if (longest < length)
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 13 of 18 · find part a in this mark scheme extract



```
                    {
                        longest = length;
                        longestWord = word;
                    }

                    if (shortest > length)
                    {
                        shortest = length;
                        shortestWord = word;
                    }
                }// end of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word

            if (count > 0)
            {
                Console.WriteLine(count + " word(s) with "+ inputWord + " in them");
                Console.WriteLine("The longest word has " + longest + " characters");
                Console.WriteLine("The shortest word has " + shortest + " characters");
                Console.WriteLine("The longest word is " + longestWord + " characters");
                Console.WriteLine("The shortest word is " + shortestWord + " characters");
                Console.WriteLine("--------------------------------------------------------------------------------");
            }
            else
            {
                Console.WriteLine("There were 0 words that had all the letters from " + inputWord + " in them");
                Console.WriteLine("--------------------------------------------------------------------------------");
            } // End of printing words
        } // End of check
    } // End of while loop
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 14 of 18 · find part a in this mark scheme extract

Java

```java
// Add your code here
Scanner input = new Scanner(System.in);
String inputWord = "";

// While there is a word run the program
while (!inputWord.equals("1")) {

    int shortest = 5000;
    int longest = 0;
    String shortestWord = "";
    String longestWord = "";

    System.out.println("Enter a word or 1 to exit ");
    System.out.println("--------------------------------------------------------------------------------");
    inputWord = input.nextLine();
    if (!inputWord.equals("1")){
        // Find words that begin with the same letter as the input word
        int count = 0;

        // Get each word in the array
        for (String word: wordArray){

            // If the first character of the word is the same as the first character of
            // the input word then print the word
            if (word.charAt(0) == inputWord.charAt(0)){
                System.out.println(word);
                count ++;
            }
        } // End of for loop

        // Print the total number of words that begin with the same letter
        System.out.println(count + " word(s) beginning with "+ inputWord.charAt(0));
        System.out.println("--------------------------------------------------------------------------------");

        // Find words that contain the input word
        count = 0;

        // Get each word in the array
        for (String word: wordArray){
            // Check to see if the word contains the inputWord
            if (word.contains(inputWord)){
                count ++;
                System.out.println(word);

                // Set the length of the word
                int length = word.length();
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 15 of 18 · find part a in this mark scheme extract



```
                    // If the length is longer than the current longest replace it and the longestWord
                    if (longest < length) {
                        longest = length;
                        longestWord = word;
                    }

                    // If the length is shorter than the shortest replace it and the shortestWord
                    if (shortest > length){
                        shortest = length;
                        shortestWord = word;
                    }
                } // End of checking word
            } // End of for loop

            // Print the number of words, the number of characters in the longest and
            // shortest word, the longest and the shortest word
            if( count > 0){
                System.out.println(count + " word(s) with "+ inputWord + " in them");
                System.out.println("The longest word has " + longest + " characters");
                System.out.println("The shortest word has " + shortest + " characters");
                System.out.println("The longest word is " + longestWord + " characters");
                System.out.println("The shortest word is " + shortestWord + " characters");
                System.out.println("--------------------------------------------------------------------------------");
            } else {
                System.out.println("There were 0 words that had all the letters from " + inputWord + " in them");
                System.out.println("--------------------------------------------------------------------------------");
            } // End of printing words
        }
    } // End of while loop
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 16 of 18 · find part a in this mark scheme extract

Python

```
# While there is a word run the program
while inputWord !="1":

    shortest = 5000
    longest = 0
    shortestWord = ""
    longestWord = ""

    inputWord = input("Enter a word or 1 to exit: ")
    print("------------------------------------------------------------------------")

    if inputWord !="1":
        # Find words that begin with the same letter as the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # If the first character of the word is the same as the first character of
            # the input word then print the word
            if word[0] == inputWord[0]:
                print(word)
                count = count + 1

        # Print the total number of words that begin with the same letter
        print("\n",count,"word(s) beginning with",inputWord[0])
        print("------------------------------------------------------------------------")

        # Find words that contain the input word
        count = 0
        # Get each word in the array
        for word in wordArray:
            # Check to see if the word contains the inputWord
            if inputWord in word:
                count = count + 1
                print(word)
                # Set the length of the word
                length = len(word)

                # If the length is longer than the current longest replace it and the longestWord
                if longest < length:
                    longest = length
                    longestWord = word

                # If the length is shorter than the shortest replace it and the shortestWord
                if shortest > length:
                    shortest = length
                    shortestWord = word
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

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```
59
60          # Print the number of words, the number of characters in the longest and
61          # shortest word, the longest and the shortest word
62          if count > 0:
63              print("\n",count,"word(s) with",inputWord,"in them")
64              print("The longest word has",longest,"letters")
65              print("The shortest word has",shortest,"letters")
66              print("The longest word is",longestWord)
67              print("The shortest word is",shortestWord)
68              print("--------------------------------------------------------------------------")
69          else:
70              print("There were 0 words with all the letters from",inputWord,"in them")
71          # ------------------------------------------------------------------------------------
72
73  # End the program when there are no more words
74  print("End of program")
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 5 of 8 · Question 2 part a · 10 marks

2021 NOV · Paper 02 · extract 18 of 18 · find part a in this mark scheme extract

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Type MS · Turn a flowchart or pseudocode into Python

Answer 6 of 8 · Question 2 part b · 11 marks

2022 JUN · Paper 02 · extract 1 of 7 · find part b in this mark scheme extract

2 (a)(i)	A1	3/8/20		(1)

Type MS · Turn a flowchart or pseudocode into Python

Answer 6 of 8 · Question 2 part b · 11 marks

2022 JUN · Paper 02 · extract 2 of 7 · find part b in this mark scheme extract



Type MS · Turn a flowchart or pseudocode into Python

Answer 6 of 8 · Question 2 part b · 11 marks

2022 JUN · Paper 02 · extract 3 of 7 · find part b in this mark scheme extract



Type MS · Turn a flowchart or pseudocode into Python

Answer 6 of 8 · Question 2 part b · 11 marks

2022 JUN · Paper 02 · extract 4 of 7 · find part b in this mark scheme extract



Type MS · Turn a flowchart or pseudocode into Python

Answer 6 of 8 · Question 2 part b · 11 marks

2022 JUN · Paper 02 · extract 5 of 7 · find part b in this mark scheme extract

2 (a)(v)	A5	pNumber		(1)

Question	mp	Answer	Additional Guidance	Mark

Type MS · Turn a flowchart or pseudocode into Python

Answer 6 of 8 · Question 2 part b · 11 marks

2022 JUN · Paper 02 · extract 6 of 7 · find part b in this mark scheme extract

2 (b)	**Award 1 mark for each of:**		Logic of algorithm must be followed as set out. Alternatives must address each point. Do not penalise candidates who attempt more than the stated requirements. Do not penalise spelling mistakes in the input message.	(11)
	Evidence should be found in the Function			
	B1	Function interface is correct (1)		
	B2	Condition correct (for the number 1) (1)		
	B3	Loop initialised correctly (from 2 to the input number) (1)		
	B4	If condition correct (modulus) (1)		
	B5	Check returned (1)		
	Evidence should be found in the Main program			
	B6	Input of number stored as an integer variable (1)		
	B7	Input includes a space/colon / new line before the input value		
	B8	Function called correctly (1)		
	B9	Display includes number and message for either a prime number **OR** not (1)		
	Overall			
	B10	Compiling without syntax errors (1)		
	B11	Executing and producing the correct output (1)		
Code examples				
Python				

```python
# Write the function here
def checkPrime(pNumber):
    if pNumber == 1:
        check = False
    else:
        check = True
        for count in range(2,pNumber):
            if pNumber % count == 0 :
                check = False
    return check

# Write the main program here
number = int(input("Enter a number: "))
result = checkPrime(number)
if result == True:
    print(number,"is a prime number")
else:
    print(number,"is not a prime number")
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 6 of 8 · Question 2 part b · 11 marks

2022 JUN · Paper 02 · extract 7 of 7 · find part b in this mark scheme extract

Question	mp		Additional Guidance	Mark

Type MS · Turn a flowchart or pseudocode into Python

Answer 7 of 8 · Question 3 part c · 5 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part c in this mark scheme extract

Question	Mp	Answer	Additional Guidance	Marks
3(a)(i)	A1 A2	Award **one** mark for each: • Pass 1 correct (1) • Pass 2 correct (1) 9 11 12 19 14 18 15 17 9 11 12 14 18 15 17 19 (1) 9 11 12 14 15 17 18 19 (1) Must show ONLY the 19 moving in first pass and ONLY the 18 moving in second pass	Ignore lines with individual swaps and only award correct completed passes 9 11 12 19 14 18 15 17 9 11 12 **14** **19** 18 15 17 9 11 12 14 **18** **19** 15 17 9 11 12 14 18 **15** **19** 17 9 11 12 14 18 15 **17** **19** (1) 9 11 12 14 **15** **18** 17 19 9 11 12 14 15 **17** **18** 19 (1) Accept **accurate** right-to-left bubble sort as shown below 9 11 12 19 14 18 15 17 9 11 12 14 19 15 18 17 (1) 9 11 12 14 15 19 17 18 9 11 12 14 15 17 19 18 9 11 12 14 18 15 17 19 (1) Complete sort cannot be the 1st /only line in response – 0 marks as this will be taken as both 18 and 19 moving in first pass	**2**

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Type MS · Turn a flowchart or pseudocode into Python

Answer 7 of 8 · Question 3 part c · 5 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part c in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark
3(a)(ii)	A3	Award **one** mark for: • 4	If right-to-left pass completed in Q3ai, can only accept 2 swaps	**1**

Question	mp	Answer	Additional Guidance	Mark
3(a)(iii)	A4	Award **one** mark for: • 3	Do not accept 2 – 3rd pass required for **algorithm** to finish If right-to-left pass completed in Q3ai, can only accept 5 passes	**1**

Question	mp	Answer	Additional guidance	Mark
3(b)	B1	Award **one** mark for: • Binary search	Accept any other divide and conquer search algorithm (google if necessary) Accept 'binary' on its own as search in question	**2**
	B2	Award **one** mark for any of the following: • Divide and conquer / list halves each time(1) • Fewer searches needed (1)	Do not accept points about being 'already sorted' as this is in question	

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Type MS · Turn a flowchart or pseudocode into Python

Answer 7 of 8 · Question 3 part c · 5 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part c in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
3(c)		Award **one** mark for each of:		
	C1	(while) Loop ends **only** when 0 entered (1)	Any method e.g. `while age != 0:`	

	C2	Avoiding divide by zero error (1)	By coding 2nd selection box from flowchart – no output required
			Must come after some attempt to process age or count – award even if inside loop.
			ignore 'other' divide by zero errors introduced by candidate
			Must follow logic of flowchart e.g.:
			`if count == 0: (or count != 0)`
			…
			`else`
			…	

	C3	Average value rounded to whole number (1)	Must come after some attempt to process age or count – award even if inside loop. Must target calculated average (even if formula not correct) Any method to get int Accept casting to int *(round not in spec, but string manipulation is)* Accept integer division	**5**

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Type MS · Turn a flowchart or pseudocode into Python

Answer 7 of 8 · Question 3 part c · 5 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part c in this mark scheme extract

	C4	Correct value calculated for average and count (1)	Allow even if 0 used to initialise count in B1 (this must be only error in calculation) – set count to -1 to test	
	C5	Count and average included in a syntactically correct final print statement (1)	Award even if never runs Must include both variables (may also include string)	

Correct

```
# Q03c

# program variables
age = 1
count = -1
total = 0

# ------------------------------------------------
# Write your code below this line

while age != 0:

    age = int(input("Please enter age in years: "))

    total = total + age
    count = count + 1

if count == 0:
    print("No ages input")
else:
    average = round(total / count)
    print("You entered", count, "ages; the average age is", average)
```

Typical

```
# Q03c

# program variables
age = 0
count = 0
total = 0

# ------------------------------------------------
# Write your code below this line
age = int(input("Please enter age in years: "))
while age != 0:

    total = total + age
    count = count + 1

    age = int(input("Please enter age in years: "))

if count == 0:
    print("No ages input")
else:
    average = int(total / count)
    print("You entered", count, "ages; the average age is", average)
```

Type MS · Turn a flowchart or pseudocode into Python

Answer 7 of 8 · Question 3 part c · 5 marks

2025 JUN · Paper 02 · extract 5 of 5 · find part c in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional Guidance	Mark

Type MS · Turn a flowchart or pseudocode into Python

Answer 8 of 8 · Question 4 part c · 8 marks

2025 JUN · Paper 02 · extract 1 of 5 · find part c in this mark scheme extract

4(a)	A1	Breaking problem into small parts		1

Question	mp	Answer	Additional guidance	Mark
4(b)		Award **two** marks for a linked explanation such as:		
	B1 B2	• Programmer can ignore unnecessary detail (1) because it doesn't matter how library generates the number (1) • Eli can focus on just his own code (1) because library hides the code that generates the random numbers (1) • It doesn't matter how the subprogram generates the random number (1) because Eli only needs the number itself (1)	First mark for meaning of abstraction (focus on necessary parts / hiding unnecessary detail) Second mark for linking use of **random** library Do **not** accept answers relating to modelling real-world scenarios as this is in the question	2

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Type MS · Turn a flowchart or pseudocode into Python

Answer 8 of 8 · Question 4 part c · 8 marks

2025 JUN · Paper 02 · extract 2 of 5 · find part c in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark
4(c)		Award **one** mark for each of:		
	C1	Booster variable initialised to 1 (1)	booster = 1	
	C2	At least one given constant used correctly (1)		
	C3	Loop used to make three throws (1)	Either type	
	C4	Use of random.randint( ) to assign roll a value between 1 and 6 (1)	roll = random.randint(1,SIDES) roll = random.randint(1,6) Any accurate method do **not** accept: random.randint(6) random.randint(SIDES)	

	C5	Check for both 3 or 6 for at least one roll (1)	any method e.g.
			`if (roll == 3 or roll == 6)`
			`if roll % 3 == 0`	

	C6	booster incremented (1)	Must be indented under condition in C5. Can award even if C5 not functioning correctly, but must be idea of only boosting on 3 and 6	
	C7	Correct final score calculated from 3 random rolls (1)	Even if not displayed Does not need to use given variable (finalScore) to get this mark	**8**

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Type MS · Turn a flowchart or pseudocode into Python

Answer 8 of 8 · Question 4 part c · 8 marks

2025 JUN · Paper 02 · extract 3 of 5 · find part c in this mark scheme extract

	C8	Correct output strings including each roll score and final score with appropriate strings (1)	All three throws and final score as shown in figure 6. `roll` and `finalScore` variables should match candidate solution Allow sensible alternate strings e.g. Roll for Rolled or Total for Score – Ignore : (colon)	

Type MS · Turn a flowchart or pseudocode into Python

Answer 8 of 8 · Question 4 part c · 8 marks

2025 JUN · Paper 02 · extract 4 of 5 · find part c in this mark scheme extract


```
# Q04c

# import libraries
import random

# program constants
SIDES = 6
THROWS = 3

# complete the lines to initialise the variables
roll = 0
subtotal = 0
booster = 1
finalScore = 0

# roll dice 3 times
for throw in range(THROWS):
    # display number rolled
    roll = random.randint(1,SIDES)
    print("Rolled:",roll)

    # update subtotal
    subtotal = subtotal + roll

    # check for roll of 3 or 6 and increment booster if needed
    if roll == 3 or roll == 6:
        booster = booster + 1

# calculate and output final score
finalScore = subtotal * booster
print("Final score:", finalScore)
```


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Type MS · Turn a flowchart or pseudocode into Python

Answer 8 of 8 · Question 4 part c · 8 marks

2025 JUN · Paper 02 · extract 5 of 5 · find part c in this mark scheme extract

Paper 2 – Application of Computational Thinking

Question	mp	Answer	Additional guidance	Mark